

AWS 3D E-Commerce Platform Architecture

Aaron Kgaphola

Scenario

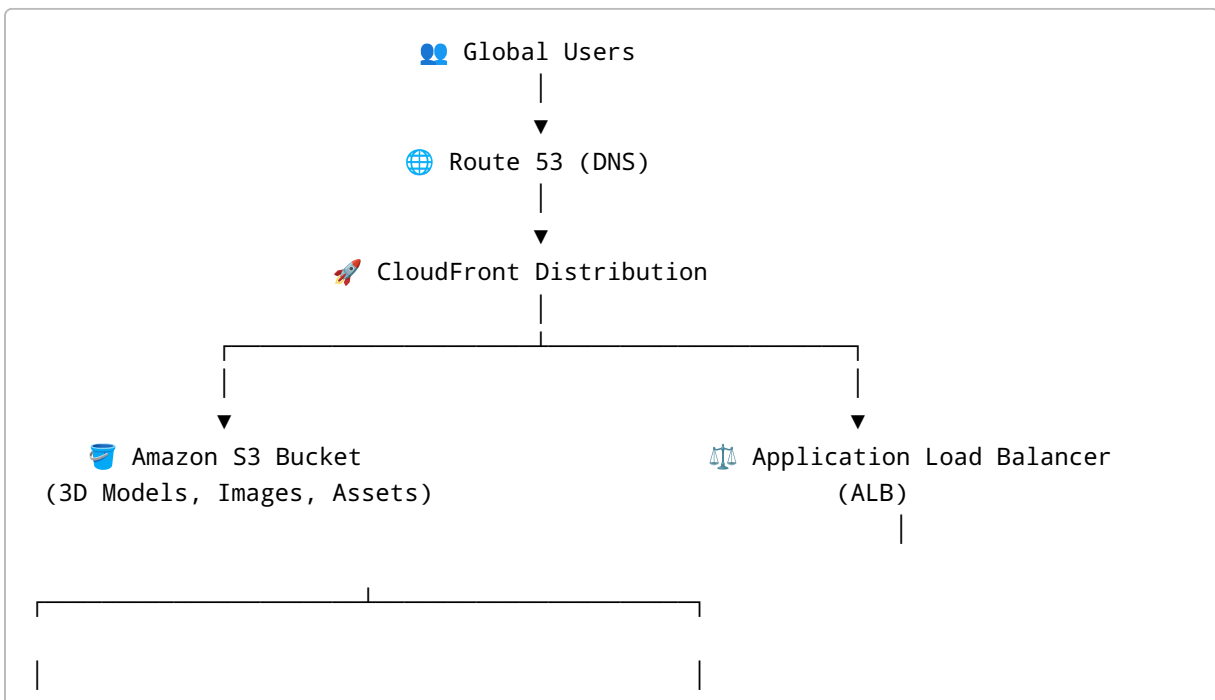
A global e-commerce company specializing in 3D products wants to migrate its platform to AWS Cloud. The company requires a highly available, scalable, secure, and cost-effective architecture capable of serving customers worldwide. The platform must efficiently store and deliver 3D models, images, and other digital assets while ensuring excellent performance and reliability during periods of high traffic.

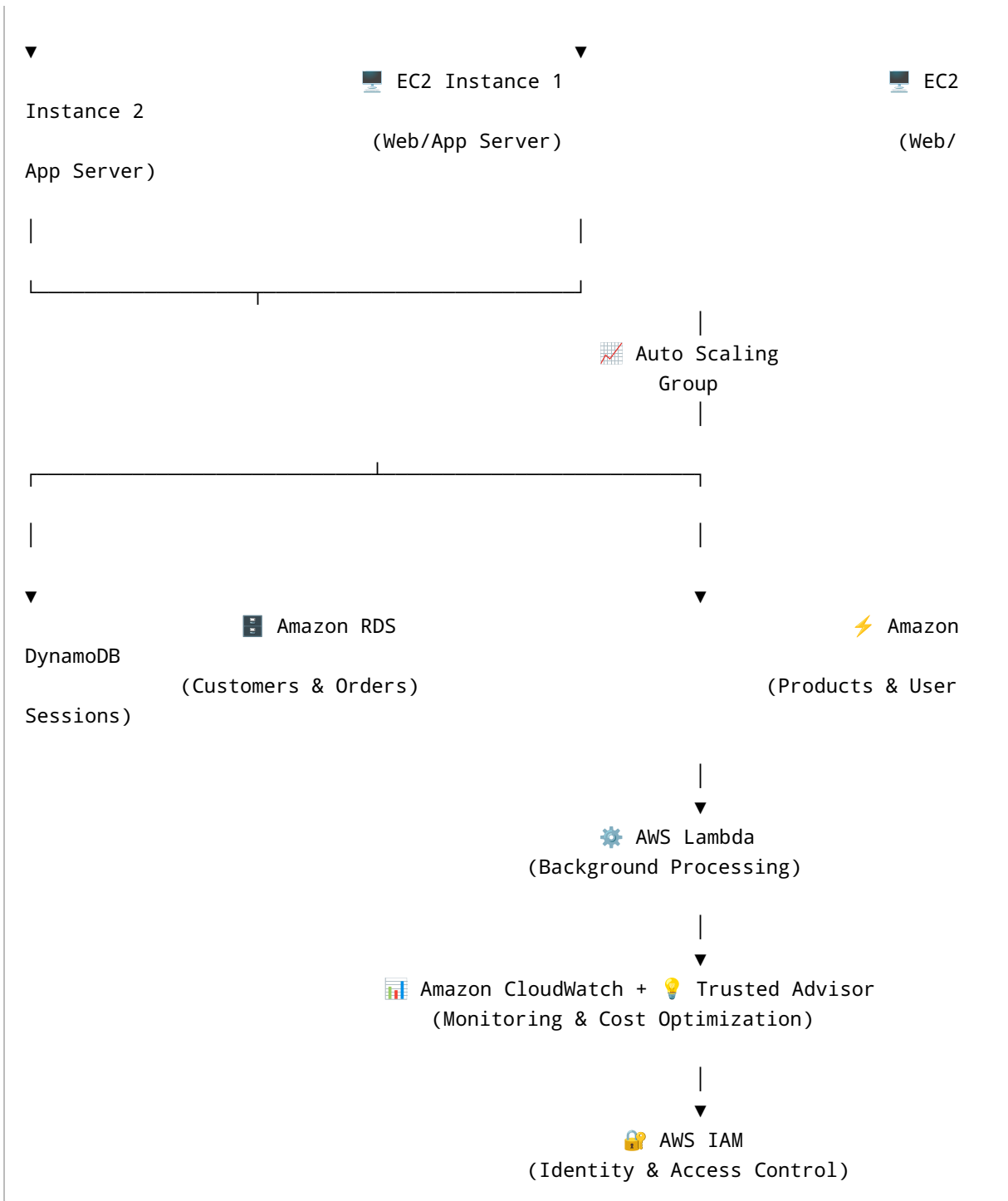
Requirements

The AWS architecture must meet the following requirements:

-  **High Availability** – Ensure the platform remains accessible even if individual components fail.
-  **Scalability** – Automatically handle increases and decreases in customer demand.
-  **Performance Efficiency** – Deliver content quickly and provide a smooth user experience globally.
-  **Security** – Protect customer data, applications, and AWS resources using best practices.
-  **Cost Optimization** – Minimize operational costs while maintaining performance and reliability.
-  **Reliability** – Ensure consistent operation and quick recovery from failures.
-  **Operational Excellence** – Monitor, manage, and continuously improve system performance.

Architecture Diagram





AWS Services Used and Why They Were Chosen

Amazon Route 53

- Provides DNS management for the application.

- Routes users to the nearest available resources.
- Supports health checks and failover capabilities.

Amazon CloudFront

- Delivers content globally with low latency.
- Caches 3D assets closer to users.
- Improves loading speed and user experience.

Amazon S3

- Stores 3D product models, images, videos, and static content.
- Highly durable and scalable.
- Cost-effective storage solution.

Application Load Balancer (ALB)

- Distributes incoming traffic across multiple EC2 instances.
- Prevents a single server from becoming overloaded.
- Improves availability and fault tolerance.

Amazon EC2

- Hosts the web application and backend services.
- Provides scalable compute resources.
- Supports customization and flexibility.

Auto Scaling Group

- Automatically adds or removes EC2 instances based on demand.
- Handles unexpected traffic spikes.
- Reduces operational costs.

Amazon RDS

- Stores customer accounts, orders, and transactional data.
- Supports Multi-AZ deployments for high availability.
- Simplifies database management.

Amazon DynamoDB

- Stores product catalog information and user session data.
- Provides fast, low-latency performance.
- Automatically scales to meet demand.

AWS Lambda

- Executes backend processes without managing servers.
- Ideal for notifications, image processing, and automation tasks.
- Cost-efficient because payment is based on usage.

AWS IAM

- Controls access to AWS resources.
- Implements least-privilege security principles.
- Enhances overall platform security.

Amazon CloudWatch

- Monitors application performance and infrastructure health.
- Generates alerts and operational insights.
- Supports proactive issue detection.

AWS Trusted Advisor

- Provides recommendations for:
 - Cost optimization
 - Security improvements
 - Performance enhancements
 - Fault tolerance best practices
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How the Architecture Meets the Requirements

High Availability

- Route 53 health checks
- Application Load Balancer
- Multiple EC2 instances
- RDS Multi-AZ deployment

Scalability

- Auto Scaling Groups
- DynamoDB automatic scaling
- CloudFront edge locations

Performance

- CloudFront content caching
- DynamoDB low-latency access
- Optimized delivery of 3D assets through S3

Security

- AWS IAM access controls
- Secure data storage
- AWS best-practice security model

✓ Cost Optimization

- Auto Scaling reduces idle resources
 - Lambda serverless processing
 - S3 low-cost storage
 - Trusted Advisor recommendations
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Designed By

Aaron Marube Kgaphola

AWS Cloud Practitioner | AWS re/Start Graduate